

RegDivergence-101: An LLM Benchmark for Cross-Jurisdiction Regulatory Contradiction Detection in Life Sciences

CHUCHU WU*, Carnegie Mellon University, USA

ZHIYIN ZHOU, Pratt Institute, USA

JINGZHUO HU, University of Pennsylvania, USA

LIANG YOU, University of Pittsburgh, USA

Pharmaceutical sponsors developing a drug for both the United States and the European Union must reconcile guidance issued independently by the FDA and the EMA. Where the two agencies require *substantively the same thing*, a sponsor can file once; where they *diverge*, a single trial design risks rejection in one region; where one agency is *silent* on a point the other regulates, the sponsor must infer obligations. Today this reconciliation is performed manually by regulatory-affairs experts. We introduce **cross-jurisdiction regulatory divergence detection**: given an FDA requirement and an EMA requirement on the same topic, classify their relationship as AGREE, DIVERGE, or SILENT. SILENT is inherently directional (SILENT_FDA vs. SILENT_EMA); we record direction per pair and report per-direction F1 alongside the collapsed label. We release **RegDivergence-101**, a 101-pair expert-grounded **pilot** evaluation benchmark (labels grounded in three peer-reviewed FDA/EMA comparison studies and primary FDA/EMA/ICH guidance text; dual-annotation inter-annotator $\kappa = 0.85$), and systematically characterise a four-method baseline hierarchy: lexical heuristic (0.511 macro-F1, 95% CI [0.411–0.605]), NLI cross-encoder (0.233), obligation-level Graph-RAG (0.663 [0.570–0.747]), and flat LLM judge / Claude Haiku (0.830 [0.747–0.908]). Three directional observations emerge at pilot scale ($n = 101$): SILENT is semantically detectable but invisible to entailment-only formulations; pair-level obligation graphs improve over lexical methods but trail flat-LLM context (CIs partially overlapping); and corpus-level graph construction is the indicated architectural target for large-scale silent-detection. RegDivergence-101 is a pilot release establishing the task formulation and baseline hierarchy; four unrepresented regulatory domains and an expansion roadmap are described in §7.

CCS Concepts: • **Computing methodologies** → **Lexical semantics**; • **Applied computing** → **Health informatics**.

Additional Key Words and Phrases: large language models, LLM benchmark, regulatory NLP, AI for compliance, FDA/EMA divergence, contradiction detection, Graph-RAG, retrieval-augmented generation, generative AI, life sciences, AGREE/DIVERGE/SILENT classification

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*Corresponding author.

Authors' Contact Information: Chuchu Wu, chuchuw@alumni.cmu.edu, Carnegie Mellon University, Pittsburgh, PA, USA; Zhiyin Zhou, zzhou20@pratt.edu, Pratt Institute, New York, NY, USA; Jingzhuo Hu, jingzhuoh00@gmail.com, University of Pennsylvania, Philadelphia, PA, USA; Liang You, liy121@pitt.edu, University of Pittsburgh, Pittsburgh, PA, USA.

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1 Introduction

Bringing a medicine to market in both the US and EU requires satisfying two regulators that publish guidance separately and frequently *differently*. Gene-therapy sponsors face a 15-year long-term follow-up recommendation from FDA while EMA defers the duration to case-by-case risk assessment—same topic, materially different specificity. Analogous divergences span pediatric extrapolation, endpoint definitions in ulcerative colitis, biosimilar comparability, and estimand implementation. A sponsor who misses one divergence during protocol design can lose a year or more re-running a study.

Reconciling the two corpora is currently a manual, expert task. Yet the NLP building blocks for automating it all exist—document-level NLI (ContractNLI [1], DocNLI [2]), legal contradiction detection (LegalWiz [3], CLAUSE [4]), and Graph-RAG (Microsoft GraphRAG [5], GraphCompliance [6], RAGulating Compliance [7]). **No prior work combines them to detect divergence between two parallel regulatory corpora from different jurisdictions.** All existing contradiction work is *intra-document* (within one contract) or *single-jurisdiction* (contract-vs-statute).

This paper makes three contributions: (1) **Task formulation** — three-way classification (AGREE / DIVERGE / SILENT) over pre-aligned FDA/EMA requirement pairs; SILENT treated as directional (SILENT_FDA / SILENT_EMA) with per-direction F1 in Table 2. (2) **Pilot benchmark** — **RegDivergence-101**, 101 pairs across 13 topic areas with a documented two-track construction protocol, $\kappa = 0.85$, and bootstrap CIs; expansion roadmap in §7. (3) **Baseline hierarchy** — four methods compared at pilot scale ($n = 101$) with bootstrap 95% CIs, yielding three directional observations on the task’s structure.

2 Related Work

Document-level NLI. ContractNLI [1] introduced document-level entailment/contradiction/not-mentioned classification with evidence extraction, and showed that contradiction detection lags entailment, especially under “negation by exception.” DocNLI [2] extended NLI to full documents.

Legal contradiction detection (2025–2026). LegalWiz [3] uses a multi-agent framework for legal contradiction detection but is confined to single-document regimes; Better Call CLAUSE [4] benchmarks 7,500+ perturbed contracts and finds LLMs struggle to justify subtle errors legally. A published RAG-robustness study directly informs our results: Yoran et al. [8] show that retrieved context which is irrelevant or non-entailing can mislead a retrieval-augmented model unless it is explicitly filtered—the failure mode in Observation 2, where modal-register similarity misleads the pair-level graph.

Graph-RAG and regulatory compliance. Microsoft GraphRAG [5] builds entity graphs with community summaries for local and global queries. GraphCompliance [6] aligns *policy graphs* (regulatory requirements) with *context graphs* (organisational facts) for GDPR; it explicitly lists “handling conflicting regulatory interpretations” as open. RAGulating Compliance [7] builds an ontology-free regulatory knowledge graph for traceable QA.

Cross-jurisdiction regulatory benchmarks. The closest concurrent work is **Sino-US-DrugQA** [9] (11,871 MCQA items, US–CN), which shares our concept-alignment thesis but cannot express the SILENT class. Comparative-regulatory-science surveys document FDA/EMA divergence across dosing [10], oncology labelling [11], and multi-country SmPCs [12], but as human-authored comparisons, not ML benchmarks. RegDivergence-101 is the first FDA/EMA benchmark with an explicit SILENT class. In high-stakes sectors such as healthcare, policy-aligned evaluation frameworks weigh safety and regulatory constraints over raw accuracy [13]; RegDivergence-101 extends this to cross-jurisdiction regulatory reasoning.

Gap. Contradiction detection is intra-document or single-jurisdiction; Graph-RAG targets single-corpus QA or single-jurisdiction compliance; FDA/EMA divergence is documented manually but never automated. Our task sits precisely in this gap.

3 Task and Dataset

3.1 Task Definition

For a topic t , let f_t be an FDA requirement statement and e_t the corresponding EMA statement. The label y_t is:

- **AGREE** — both jurisdictions require substantively the same thing (same modal strength, same threshold, same scope of applicability). Paraphrase equivalence and synonymous modals (*shall* / *must*) are treated as AGREE; minor drafting differences that do not alter sponsor obligations are disregarded.
- **DIVERGE** — the requirements directly conflict: they impose different thresholds, different modal strengths (*shall* vs. *should*), different timelines, or mutually incompatible design specifications.
- **SILENT** — one jurisdiction does not address what the other regulates. SILENT is *directional*: **SILENT_FDA** (FDA silent, EMA regulates) vs. **SILENT_EMA** (EMA silent, FDA regulates). Direction is recorded in the `silent_direction` field and reported in Table 2; a unified SILENT label is retained in Table 1 for backward compatibility.

Epistemic status of SILENT labels. A SILENT label is an absence claim: it asserts that no counterpart requirement was found after an exhaustive corpus search. This carries irreducible epistemic uncertainty. We bound this uncertainty through (a) a documented corpus-search protocol (§3.3) and (b) a per-pair `doc_confidence` flag (HIGH / MODERATE). Downstream work should treat MODERATE-confidence SILENT pairs as soft labels.

Task scope. This formulation assumes requirement pairs are *pre-aligned*—both sides address the same regulatory topic, established via the two-track protocol (§3.2). Topic alignment is a non-trivial upstream problem treated separately; the corpus-level architecture that would subsume it is described in §7.

3.2 Pair Construction Protocol

RegDivergence-101 is constructed via two tracks that differ in how topical correspondence between FDA and EMA requirements is established. Per-pair source-track attribution (`track`: A or B) is released with the dataset.

Track A — expert-comparison-derived pairs (59 pairs). Extracted from three open-access peer-reviewed studies that comparatively mapped FDA and EMA requirements: Vieujean et al. [14] (ulcerative colitis, 40 pairs), Boesen et al. [15] (psychiatric drug trials, 10 pairs), and Schwarz and Decristoforo [16] (radiopharmaceuticals, 9 pairs). Domain experts in those studies had already established co-topicality; our task was to classify the AGREE/DIVERGE/SILENT relationship between their pre-aligned pairs.

Track B — primary guidance pairs (42 pairs). For pairs from ICH E9(R1), E11A [17], and FDA/EMA biosimilar and gene-therapy guidance, co-topicality was established by (1) anchoring requirements to a shared ICH Common Technical Document (CTD) module/sub-topic, then (2) expert confirmation that the candidate pair was genuinely co-topical.

Specificity exclusion criterion. Candidate pairs operating at different regulatory hierarchy levels (e.g., *statistical analysis plan* vs. *general statistical principles*) were excluded to prevent conflating level-of-detail differences with substantive DIVERGE labels. Excluded pairs and the full exclusion-criteria checklist are released in `excluded_pairs.jsonl` for third-party audit.

Each pair records {id, topic, track, source, fda_text, ema_text, label, silent_direction, doc_confidence, disputed, annotation_notes}. The combined release contains **101 pairs**: 42 AGREE, 38 DIVERGE, 21 SILENT, across 13 topic areas. We report macro-F1 to weight all classes equally.

3.3 Annotation Protocol

Annotators. Annotator A (C. Wu) is a solution architect for a regulatory-document AI platform; Annotator B (Z. Zhou) is a product manager for an AI-powered clinical-trial management platform. Neither holds formal RA credentialing; Track A expert-panel papers serve as the external human-expert anchor.

Stages. Stage 1: Wu assigned labels via definitions-based framing; Stage 2: Zhou independently re-labelled all 101 pairs under a sponsor-obligation framing to probe framing-independent stability. The κ is a cross-framing consistency measure, not identical-instruction IAA. An independent third-annotator pass is planned for RegDivergence-500.

IAA. Human cross-framing $\kappa = 0.85$ (101 pairs, observed 0.90)—*substantial* per Landis and Koch [18], on par with MedNLI (0.65) and ContractNLI (≈ 0.72). SILENT near-perfect ($\kappa = 0.98$); AGREE/DIVERGE more variable (≈ 0.79 –0.81). A secondary LLM consistency check (36-pair sample) yielded $\kappa = 0.542$ (test-retest, not human IAA).

Disagreements (n = 10). Six resolved by source-paper language; four genuinely borderline pairs retain the Stage 1 label and are flagged disputed = True in the dataset with written justifications.

SILENT confidence. All 21 SILENT pairs carry doc_confidence = MODERATE: exhaustive corpus search was performed (full FDA and EMA guidance, Q&As, reflection papers) but absence claims remain epistemically soft. The HIGH/MODERATE audit is deferred to RegDivergence-500.

4 Baselines

4.1 Evaluation Protocol

RegDivergence-101 is an **evaluation benchmark**, not a training dataset. All baselines are zero-shot or few-shot; results are reported as a ranking trend ($\Delta = 0.15$ –0.60 macro-F1 between adjacent methods, CIs partially overlapping), not pairwise significance.

All methods use **stratified 5-fold splits** (5 seeds) with **bootstrap 95% CI** (1,000 resamples). Lexical and NLI methods use **nested CV** (inner folds tune thresholds); Graph-RAG and LLM judge have no tunable hyperparameters (LLM judge scored on all 101 pairs, bootstrap CI only).

4.2 Lexical Heuristic

No neural model, no API—fully offline and deterministic. Combines TF-IDF cosine similarity with five conflict signals: negation asymmetry, divergence cue pairs (e.g., *single-arm* vs. *randomised*), modal-strength asymmetry (*must/shall* vs. *may/encouraged*), numeric-threshold mismatch, and one-sided specificity (concrete number vs. case-by-case deferral). A shared ICH citation is a strong AGREE signal.

4.3 NLI Cross-Encoder

typeform/distilbert-base-uncased-mnli, run in both directions; labels derived from contradiction/neutral/entailment probabilities with CV-tuned thresholds. Any NLI model lacking an explicit SILENT class maps absence-of-regulation to *neutral*, structurally collapsing SILENT recall (Observation 1 in §6); stronger encoders improve AGREE/-DIVERGE F1 but cannot resolve this gap without task-specific supervision.

Table 1. Macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for the four baselines ($n = 101$ pairs).

Method	Macro-F1 [95% CI]	Acc	AGREE F1 [CI]	DIVERGE F1 [CI]	SILENT F1 [CI]
Lexical heuristic ¹	0.511 [0.411–0.605]	0.541	0.718 [0.605–0.809]	0.466 [0.320–0.593]	0.334 [0.133–0.537]
NLI cross-encoder ²	0.233 [0.152–0.303]	0.238	0.279 [0.103–0.449]	0.383 [0.247–0.506]	0.025 [0.000–0.086]
Graph-RAG pair-level ³	0.663 [0.570–0.747]	0.673	0.754 [0.646–0.851]	0.586 [0.433–0.722]	0.640 [0.478–0.771]
LLM judge (Haiku)⁴	0.830 [0.747–0.908]	0.842	0.905 [0.831–0.965]	0.805 [0.696–0.892]	0.778 [0.609–0.914]

¹Nested 5-fold CV, 5 seeds; thresholds tuned on inner folds only. ²Same CV protocol, thresholds tuned on inner folds. ³No tunable hyperparameters; bootstrap CI only. ⁴claude-haiku-4-5-20251001, default temperature; bootstrap CI only. Estimated cost: ~\$0.10 for all 101 pairs.

4.4 Graph-RAG Pair-Level Classifier

For each text, Claude Haiku extracts a policy triplet $\langle \text{subject}, \text{obligation_level}, \text{requirement}, \text{conditions} \rangle$ where $\text{obligation_level} \in \{\text{MANDATORY}, \text{PROHIBITED}, \text{RECOMMENDED}, \text{PERMITTED}, \text{SILENT}\}$. Only MANDATORY $\leftrightarrow$ PROHIBITED is a hard-coded DIVERGE rule; all other pairs are escalated to a second LLM call that reasons over the *obligation structure* of the aligned node pair, with explicit calibration that modal-register differences (RECOMMENDED vs. MANDATORY) do not automatically imply DIVERGE.

Within-model scope. Both the Graph-RAG classifier and the flat LLM judge use claude-haiku-4-5-20251001. The comparison between them is therefore a *within-model ablation*—with vs. without explicit obligation-graph structure—not an architectural comparison across model families. Conclusions about the relative value of graph structure should be confirmed with a second model family before generalisation.

4.5 LLM Judge (Flat Pairwise)

Claude Haiku with explicit AGREE / DIVERGE / SILENT definitions, prompted once per pair with both texts. No graph, no retrieval. Full prompt and code at Appendix A.

5 Results

Table 1 reports macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for all four methods.

Table 2. Per-direction SILENT F1 for all four methods (bootstrap 95% CI, $n = 101$ pairs). SILENT_FDA ($n = 13$): FDA is silent, EMA regulates. SILENT_EMA ($n = 8$): EMA is silent, FDA regulates.

Method	SILENT_FDA F1 [95% CI]	SILENT_EMA F1 [95% CI]
Lexical heuristic	0.277 [0.000–0.545]	0.393 [0.000–0.706]
NLI cross-encoder	0.049 [0.000–0.167]	0.000 [0.000–0.000]
Graph-RAG pair-level	0.716 [0.516–0.882]	0.535 [0.267–0.762]
LLM judge (Haiku)	0.795 [0.588–0.952]	0.738 [0.429–0.952]

SILENT-FDA: 13 pairs (FDA silent, EMA active). SILENT-EMA: 8 pairs (EMA silent, FDA active). Wide CIs on SILENT-EMA reflect the small stratum ($n = 8$). Bootstrap 1,000 resamples; stratified by 3-class label.

Point estimates increase monotonically lexical $\rightarrow$ Graph-RAG $\rightarrow$ LLM judge; bootstrap CIs partially overlap at $n = 101$, so we treat this as a ranking trend rather than a significance claim. **Lexical ceiling:** errors share the signature of *high lexical overlap, opposed regulatory stance* (e.g., FDA “modified Mayo score of 5–9” vs. EMA “full Mayo score of 9–12”; same vocabulary, different instrument), requiring semantic understanding of what each requirement asserts.

6 Analysis

6.1 What the Baseline Hierarchy Reveals

Observation 1 – SILENT is semantically detectable but invisible to NLI framing. SILENT F1 = 0.025 under NLI (no native SILENT class) vs. 0.778 under the LLM judge (explicit definition). This contrast reflects a structural constraint—the SILENT class requires a formulation that can express absence, which NLI entailment cannot—rather than a model-capacity effect. A higher-capacity NLI encoder would improve AGREE/DIVERGE F1 but would not resolve the absence-detection gap without task-specific supervision.

Observation 2 – For Claude Haiku on this dataset, pair-level obligation graphs improve over lexical (+0.15 F1) but trail flat prompting (−0.17 F1); CIs partially overlap at pilot scale. The graph makes obligation structure explicit, helping distinguish stance differences from similar-sounding text. But it discards full-text context that flat prompting uses to separate *same-requirement* / *different-modal-register* (AGREE) from *same-topic* / *different-threshold* (DIVERGE): gene-01 (FDA 15 years vs. EMA case-by-case) both extract as RECOMMENDED and the graph calls them AGREE; the flat LLM correctly identifies the threshold conflict from full text [8]. This comparison is a within-model ablation (graph vs. no-graph, both using Haiku), not a cross-model claim.

Observation 3 – Two distinct problem scopes suggest two distinct architectural targets. The *pair-level scope* (given pre-aligned sentences, label AGREE/DIVERGE/SILENT) is largely addressed by the flat LLM judge (0.830). The *corpus-level scope* (“is EMA silent anywhere across its corpus?”) requires finding aligned pairs, not just classifying them; a corpus-level policy graph is needed. §7 motivates this architecture as the next contribution.

6.2 Source-Stratified Performance Analysis

To assess whether performance reflects source-study-specific patterns rather than genuine task competence, we analyse performance stratified by track and by source concentration.

Track-level. On AGREE/DIVERGE classes, the LLM judge scores *lower* on Track A (0.769/0.745) than Track B (0.966/0.923): exposure to source studies does not inflate performance. The macro-F1 gap (Track B 0.630 vs. Track A 0.765) is explained by SILENT support ($n = 0$ vs. $n = 21$), not memorisation.

Source-concentration. 40/101 pairs come from one source (Vieujean et al.). LLM judge on the JCC stratum ($n = 40$): 0.804; on all other pairs ($n = 61$): 0.844. Performance is marginally *higher* outside JCC, ruling out domain-concentration inflation.

7 Future Work

Corpus-level policy graph. The correct long-term target is corpus-level silent detection: (1) build per-jurisdiction policy graphs indexing all FDA/EMA guidance; (2) align requirement nodes via entity normalisation and embedding similarity; (3) classify aligned pairs as AGREE/DIVERGE/SILENT by obligation-structure comparison; (4) generate natural-language explanations. This architecture subsumes the pair-alignment assumption: alignment becomes a graph-construction step, not a pre-processing input.

RegDivergence-500. Four domains are unrepresented in the pilot: CMC, adaptive trial design/estimands, statistical analysis plan requirements, and labelling/SmPC. RegDivergence-500 (≈ 500 pairs) will cover all four via a community-annotation pipeline seeded from a ~ 45 -source FDA/EMA comparison inventory. Annotation guidelines are released

with this paper. Multi-model replication, an independent third-annotator validation pass, and a cross-pair generalisation probe against Sino-US-DrugQA [9] are planned; a third jurisdiction (PMDA) will add an ≈ 40 -pair FDA $\times$ PMDA subset [10, 11].

8 Limitations

Scale and alignment. 101 pairs from three source studies limits statistical power to directional observations. Track B co-topicality judgements are single-annotator; Track A inherits source-study alignment methodology. Reproducibility depends on the released exclusion-criteria checklist (data/alignment_protocol.md). Oncology, rare diseases, medical devices, and vaccines are absent; RegDivergence-500 (§7) is required before domain-general conclusions.

SILENT labels. All 21 SILENT pairs carry doc_confidence = MODERATE; treat them as soft labels. The pair-level SILENT task is a proxy for the correct corpus-level formulation (§7).

Single-model evaluation. All LLM-dependent components use Claude Haiku. The Graph-RAG vs. LLM comparison is a within-model ablation, not a cross-model comparison. The $\kappa = 0.542$ LLM consistency check reflects the same model’s label tendencies. Haiku may have seen FDA/EMA guidance during pretraining; the source-stratified analysis (§6.2) provides evidence against inflation, but future work should evaluate on post-cutoff guidance. Multi-model replication is deferred to RegDivergence-500.

Use norms. RegDivergence-101 is a pilot benchmark: not sufficient for deployment decisions, not domain-generalizable without held-out evaluation, and not a sole criterion for AI procurement in regulatory-affairs workflows. Conclusions are specific to the FDA/EMA pair until cross-pair generalisation probes are run.

9 Conclusion

We introduced cross-jurisdiction regulatory divergence detection and released **RegDivergence-101**, a 101-pair expert-grounded **pilot** benchmark ($\kappa = 0.85$, two-track construction protocol, per-direction SILENT annotation) with a four-method baseline hierarchy. Three directional observations at pilot scale: SILENT requires explicit absence-aware formulation (NLI F1 0.03 $\rightarrow$ LLM F1 0.78); for Claude Haiku, pair-level graph structure improves over lexical (+0.15 F1) but trails flat prompting (−0.17 F1); and the pair-level vs. corpus-level scope distinction motivates the corpus-level graph architecture (§7) as the next contribution. RegDivergence-500 will establish the task at production scale.

Acknowledgments

This work was conducted independently. Benchmark pairs were derived from publicly available FDA/EMA guidance documents and open-access peer-reviewed comparison studies. LLM API calls used the Anthropic API (Claude Haiku, claude-haiku-4-5-20251001).

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A Prompts and Code

Full prompts (LLM judge system/user, Graph-RAG Stage 1 triplet-extraction, Stage 2 graph-classification), evaluation scripts, and dataset schema are available at the project repository: <https://github.com/chuchugo/cross-jurisdiction-regulatory-divergence>

LLM judge (`llm_judge.py`): system prompt defines AGREE/DIVERGE/SILENT with explicit absence framing; user prompt pairs `{fda_text}` and `{ema_text}`. Model: `claude-haiku-4-5-20251001`, temperature 1.0, max tokens 100.

Graph-RAG (`graph_rag.py`): Stage 1 extracts a policy node `{subject, obligation_level, requirement, conditions}`; Stage 2 classifies the aligned node pair. Hard rules: MANDATORY $\leftrightarrow$ PROHIBITED $\Rightarrow$ DIVERGE (no LLM call); both SILENT $\Rightarrow$ SILENT. Max tokens: 300.